

Playing with the Rubik's Cube and its God's number in Rocq

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Abstract. God's number, the largest number of face turns needed to solve a Rubik's cube, is twenty. It was settled in 2010 by a computation of thirty-five processor years. Even though this kind of computation cannot easily be replicated in a proof assistant like Rocq, three smaller results about the Rubik's cube are proved here. First, we prove that twenty is a lower bound: one position, the superflip, cannot be solved in nineteen. A half turn can also count as two moves. The number is then twenty-six. We prove that twenty-six is a lower bound for solving the four-spot with the superflip on it. This is our second result. Finally, the published computation splits the cube into the 2 217 093 120 cosets of a subgroup, one search to a coset. A coset contains 19 508 428 800 positions. We formalise the correctness of a coset search, and we apply it to one specific coset. It follows that every position of the superflip's coset is solved in twenty moves or less. This is our last result.

Keywords. Rubik's cube, God's number, formal proof, Rocq, group theory.

1 The problem

A Rubik's cube is built from twenty-six small cubes: **eight corner pieces** with three stickers each, **twelve edge pieces** with two, and **six centre pieces** with one. The centres are attached to the core. They spin in place but never travel, so they fix the frame. The white face is wherever the white centre is. A face turn moves four corners and four edges, and nothing else.

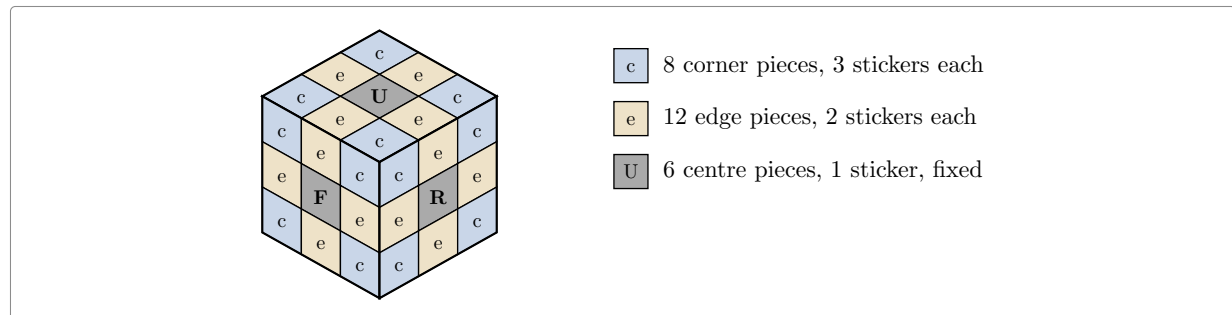


Figure 1: The three kinds of piece. Every face shows four corner stickers, four edge stickers and one centre.

Not every arrangement of the pieces can be reached by turning faces. The number of arrangements that can be reached is

$$8! \cdot 3^7 \cdot 12! \cdot 2^{11} / 2 = 43\,252\,003\,274\,489\,856\,000 \approx 4.3 \cdot 10^{19},$$

This number is read as follows. The eight corners can be in any order ($8!$). Each is twisted one of three ways, but the last one is forced by the other seven (3^7). The twelve edges can be in any order ($12!$). Each is flipped or not, and again the last one is forced (2^{11}). The result is then halved, because corners and edges cannot be rearranged independently of each other.

Turning one face is a *move*, and a half turn counts as one move just like a quarter turn. Every position can be solved. The question is how many moves the worst position needs. That number is called **God's number**.

Counting a half turn as one move is a choice. A half turn can also count as two moves, and that gives a second number for the same cube. We prove a lower bound for each.

In 2010 Rokicki, Kociemba, Davidson and Dethridge showed that it is **20** [1]. Twenty moves always suffice, and twenty moves are sometimes needed. The first half is the huge computation, and the last section of this note says how it was obtained. For that it is enough to take one position and show it cannot be solved in 19.

We take one position: the **superflip**, drawn in Figure 2 beside a solved cube. Every corner sticker is where it belongs. Every edge is in its own place but turned over, so it shows the colour of the face beside it. Look at the cube from any angle, or in a mirror, and the pattern is the same. The superflip is one of the rare positions that all 48 ways of looking at a cube leave unchanged, and that matters later. A 20-move solution for it is known, so ruling out a 19-move solution puts the superflip at distance exactly 20. We prove this bound.



Figure 2: A solved cube, and the superflip.

That is still a big computation. There are 18 moves at each step, so 19 moves means about 18^{19} words.

Three key facts about a position are used in what follows.

- Each corner has exactly one sticker of the top colour or the bottom colour. That sticker can be in three places on the corner. It can be on the top or bottom face, which we write 0, or on one of the corner's two sides, which we write 1 and 2. We call this the corner's **twist**.
- Each edge has a right way round, which we write 0. Put back the other way round, it shows its two colours the wrong way about, and that we write 1. We call this the edge's **flip**.
- Four of the twelve edges belong in the middle layer, between the top and the bottom face. We call that layer the **slice**. The four edges can sit in $\binom{12}{4} = 495$ sets of four slots. We number these sets 0 to 494. Number 0 is the slice itself.

The three are easy to read on the superflip. Every corner is home and the right way up, so every top or bottom sticker is on the top or bottom face and its eight twists are all zero. Every edge is turned over, so its twelve flips are all one. Every edge is also in its own slot, so the four middle edges are back in the four slots of the slice, which is 0.

2 The cube as permutations

The cube is easier to reason about if we stop treating it as a solid object. Only the coloured stickers matter. There are six faces of nine stickers, and the six centre stickers never move relative to each other. So a move is a rearrangement of the **48 remaining stickers**. We number them 0 to 47, as in Figure 3 and Figure 4. Each face has eight of them, taken left to right and top to bottom, with the centre skipped. Up gets 0–7, left 8–15, front 16–23, right 24–31, back 32–39 and down 40–47. The sources use these numbers, so a move can be checked against a picture.



Figure 3: Three of the six faces, and the turn of the top one.



Figure 4: The cube unfolded, with all forty-eight places numbered.

With the places numbered, a move is written down by saying where the sticker in each place goes. Turn the top face clockwise. The sticker in corner 0 goes to corner 2, the one in 2 to 7, the one in 7 to 5, and the one in 5 back to 0. That is a four step cycle, written $(0\ 2\ 7\ 5)$. The four edge stickers of that face do the same, $(1\ 4\ 6\ 3)$. The turn does not only move the top face. It also carries the top row of each side face round to the next one: front to left, left to back, back to right, right to front. That is three more cycles, $(8\ 32\ 24\ 16)$ and its two companions. Figure 5 shows all of it, each square saying which sticker sits there afterwards. The top row of the left face holds 16, 17, 18, the stickers that came round from the front. The other forty stickers stay where they are, and the six centres never move.



Figure 5: Everything a clockwise turn of the top face moves.

Six clockwise quarter turns generate everything: up, right, front, down, left and back. Each of them can also be done twice or backwards, which gives the **eighteen moves**

U	R	F	D	L	B
U2	R2	F2	D2	L2	B2
U'	R'	F'	D'	L'	B'

A move turned twice is written U2, and a move turned backwards U'. A position is a product of moves, for instance R U R' U', a word of length 4. The set of all positions is a group G : the **cube group**. Solving a position in d moves means writing it as a word of d moves. So “solvable in at most d moves” says that the position lies in the **ball of radius d** around the solved cube. God’s number is the largest distance that occurs.

2.1 The cube in Rocq

The development is written in the Rocq prover [2] and built on **mathcomp** [3], a large library of formalised mathematics that already has permutations, groups and products. The cube file, `Rubik333.v`, is a transcription of the paragraphs above, and it is short:

```
Definition facelet := 'I_48.
```

```
Definition Umove : {perm facelet} := cyc [:: 0@; 2@; 7@; 5@] * cyc [:: 1@; 4@;
6@; 3@] * cyc [:: 8@; 32@; 24@; 16@] * cyc [:: 9@; 33@; 25@; 17@] * cyc [:: 10@;
34@; 26@; 18@]. (* ... and five more, one per face ... *)
```

```
Definition faces : seq {perm facelet} :=
[:: Umove; Rmove; Fmove; Dmove; Lmove; Bmove].
```

```
Definition moves : seq {perm facelet} :=
flatten [seq [:: g; g ^+ 2; g ^-1] | g <- faces].
```

```
Definition G : {group {perm facelet}} := <<Sset>>.
```

Line by line:

- 'I_48 is the type of the whole numbers **below** 48, so the places are numbered **0 to 47** and not 1 to 48, everywhere in the sources and in the pictures of this note.
- {perm facelet} is the type of **permutations** of those places: a way of sending each place to a place, no two of them landing on the same one. That is exactly what a position is.
- cyc [:: 0@; 2@; 7@; 5@] is the **cycle** that sends 0 to 2, 2 to 7, 7 to 5 and 5 back to 0, leaving the other forty-four places where they are. The @ is local notation turning a plain number into a place.
- * composes two permutations, so Umove is the five cycles of Figure 5 done together, and g ^+ 2 and g ^-1 are the same turn done twice and undone. Its order is the opposite of the usual one.

Mathcomp applies permutations on the right, so $(g * m) f$ is $m (g f)$: a product reads left to right, like a sequence of moves played one after the other.

- **seq** is a list, and **faces** is the list of the six clockwise quarter turns. **moves** runs through it and keeps three moves per face, which is the eighteen.
- **<<Sset>>** is the group generated by a set: everything reachable by composing moves, which is the cube group.

The superflip is written down the same way in **Diameter.v**, as the twelve swaps that exchange the two stickers of each edge:

```
Definition Spcyc : seq (seq facelet) :=
  [:: [:: 1@; 33@]; [:: 3@; 9@]; [:: 4@; 25@];
    (* ... nine more, one per edge ... *) ].
```

```
Definition superflip : {perm facelet} := \prod_(l <- Spcyc) cyc l.
```

That makes it a permutation of the stickers, but on its own it says nothing about the cube. A permutation is a legal position only if the faces can be turned to reach it, that is, only if it lies in G . The definition above does not give that. It has to be proved.

The proof is one equality. On the left, the superflip as just defined. On the right, a word of twenty moves:

$$U R^2 F B R B^2 R U^2 L B^2 R U^{-1} D^{-1} R^2 F R^{-1} L B^2 U^2 F^2$$

Both sides are permutations of the 48 stickers. Each is written out as the list of the 48 places it sends each place to, so the equality is one comparison of two lists. Every letter on the right is one of the eighteen moves, so the superflip lies in G . That same word gives the upper bound of 20 for this one position.

Nothing here is assumed. There is no axiom saying what a cube is. A reader who wants to check the model has only to compare the six lists of cycles against Figure 4. After these two files, stickers are never mentioned again.

3 Searching for the lower bound

We want to know that the superflip cannot be solved in nineteen moves. The naive way is to try every word of at most nineteen moves, starting from the superflip: if the solved cube never turns up, the lower bound is twenty. A tree of depth 19 with a branching of 18 is 18^{19} words, far too many to make the search practical. This naive method has to be refined.

3.1 The pruning estimate

A first refinement is a test that cuts in advance the branches which cannot succeed. For that we use at each position a cheap lower bound h on the number of moves still needed. If h is 20 and only 18 moves remain, the branch cannot reach the solved cube in time. It is cut, with everything below it, as Figure 6 shows.

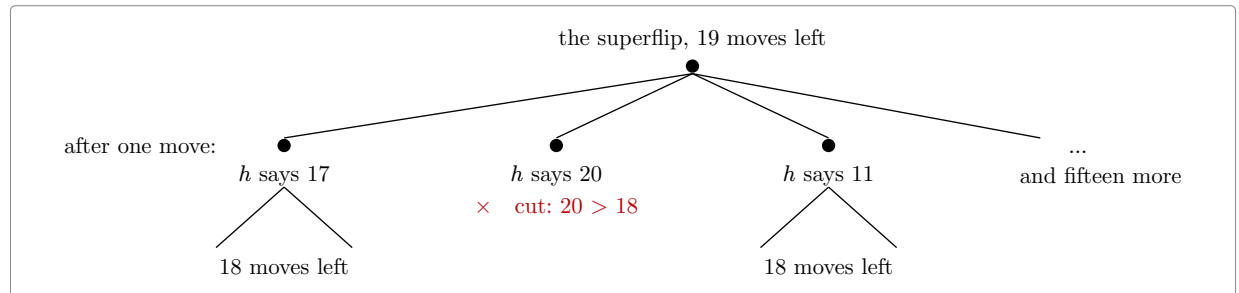


Figure 6: The search, and its scissors.

Such a lower bound is called an **admissible** estimate, and a depth-first search that deepens step by step and prunes on one is Korf’s IDA* [4].

3.2 How to get a cheap estimate

The estimate has to come from somewhere, and the idea is to forget most of the cube. Keep only part of the information, say how the corners are twisted and where the four middle-layer edges sit, and call what is left a **summary**. Many positions can share the same summary. Moves act on summaries as well as on cubes. Summaries are few, so we can compute the exact distance of each one to the solved summary and keep them all in a table. This gives *h*: take a position, compute its summary, and look its distance up in the table. This idea of summary comes from Culberson and Schaeffer, who call such a table a *pattern database* [5], and Korf solved the cube optimally with three of them [6].

The summary we have used is Kociemba’s, from his two-phase solver [7]. Figure 7 shows the three things it records. A corner has one sticker belonging to the up or down face, and that sticker sits in one of three places, which is 0, 1 or 2. An edge is either the right way round or turned over, which is 0 or

1. The four edges of the middle layer occupy four of the twelve edge slots. The

figure shades them on the two visible faces, where three of the four can be seen. Nothing else about the position is recorded.

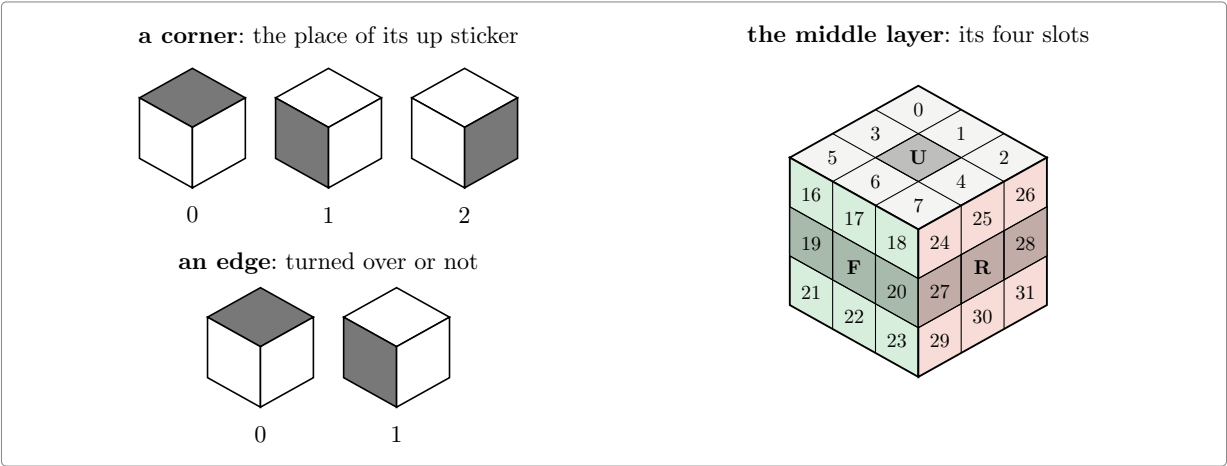


Figure 7: The three things a summary records.

The summary is the product of the three values:

summary	values	
how the eight corners are twisted	2 187	= 3 ⁷
how the twelve edges are flipped	2 048	= 2 ¹¹
where the four middle-layer edges sit	495	4 places among 12
the summary, all three together	2 217 093 120	

Every summary stands for exactly 19 508 428 800 real positions. The table records, for each summary, its distance from the solved summary. Four bits hold one entry and the whole table is **1.18 GB**. The cut is quite effective. A search at depth 14 visits 470 786 nodes. Without the cut the same tree holds $1.07 \cdot 10^{15}$ of them, so the search sees one node in two billion. We call this summary the **phase 1 summary**, after the first phase of Kociemba’s solver [7], and its table of distances the **phase 1 table**.

4 The search in Rocq

4.1 The generic search

The search at depth d is implemented in Rocq by a generic search, `Search.v`, about a hundred lines that never mention the cube. It is given a group, a set of moves and an estimate h , with only two assumptions attached to h :

Hypothesis `h1` : $h\ 1 = 0$.

Hypothesis `hstep` : $\text{forall } g\ m, m \in S \rightarrow h\ g \leq (h\ (g * m)).+1$.

Fixpoint `search` ($d : \text{nat}$) ($g : gT$) : $\text{bool} := (h\ g \leq d) \ \&\& \ ((g == 1) \ || \ (\text{if } d \text{ is } d'.+1 \text{ then has } (\text{fun } m \Rightarrow \text{search } d' \ (g * m)) \ \text{Sseq} \text{ else false}))$.

Corollary `searchN` $d\ g$: $\text{search } d\ g = \text{false} \rightarrow g \notin \text{ball } S\ d$.

Again, line by line:

- `gT` is the group the file works in. It is a variable, so nothing here is about the cube, which is supplied later at instantiation.
- `1` is the unit of that group, which for the cube is the **solved** position. So $h\ 1 = 0$ says the estimate of a solved cube is zero, and $g == 1$ asks whether the search has arrived.
- `Sseq` is the list of moves, the eighteen of them, in the order the search walks over them. `S` is the same thing seen as a set.
- $g * m$ is the position g followed by the move m , in the left to right order met above.
- $h\ g \leq d$ is the cut, and $(h\ (g * m)).+1$ is the estimate after a move plus one, which is the assumption that one move changes the estimate by at most one.
- `has (fun m => search d' (g * m)) Sseq` tries every move with one fewer move available, and answers as soon as one of them succeeds.

The last line exactly states what we need about the search: **if the search returns false, the position is not in the ball of radius d .**

4.2 The summary and its table

The estimate h is built in a second generic file, `Coord.v`. It is given the summary of a position, the way a move acts on a summary, and the table of distances. Three hypotheses come with them, one about the summary and two about the table.

Variable `coord` : $\{\text{perm facelet}\} \rightarrow X$.

Variable `act` : $X \rightarrow \{\text{perm facelet}\} \rightarrow X$.

Hypothesis `coordM` : $\text{forall } g\ m, \text{coord } (g * m) = \text{act } (\text{coord } g)\ m$.

Variable `D` : $X \rightarrow \text{nat}$.

Hypothesis `D0` : $D\ (\text{coord } 1) = 0$.

Hypothesis `Dstep` : $\text{forall } x\ m, m \in \text{Sset} \rightarrow D\ x \leq (D\ (\text{act } x\ m)).+1$.

X is the type of summaries and `coord` gives the summary of a position. `act` plays a move on a summary directly, without going back to the position it came from. `coordM` says that the two agree: playing a move and then summarising gives the same answer as `act`.

The `search` of `Search.v` is refined to carry the summary beside the position, updating it with `act` at each move. The position is still carried, since only it tells whether the cube is solved. The refined search also plays fewer moves. Turning the same face twice running is never useful, and the next subsection gives the other rules of this kind. Below is its shape, with simplified names and some details left out. The real one is `searchz3` in `Farp1.v`:

Fixpoint `search` ($d : \text{nat}$) ($g : gT$) ($x : \text{summary}$) ($p : \text{move}$) : $\text{bool} :=$
 $(D\ x \leq d) \ \&\&$

```

((g == 1) ||
 (if d is d'.+1
  then has (fun m => search d' (g * m) (act x m) m) (allowed p)
  else false)).

```

Two things travel down the tree instead of one, and each is used for exactly one job:

- $D \times \leq d$ is the cut. It reads the table at the summary x , and never looks at the position.
- $g == 1$ asks whether the cube is solved. It reads the position g , and never looks at the summary.
- $g * m$ moves the position and $act \ x \ m$ moves the summary, side by side, one move at a time. That step is `coordM` being used, and it is why the summary never has to be recomputed from the position.
- p is the move just played, and `allowed p` is the list of moves the rules permit after it. That is where redundant sequences are dropped.

4.3 The table and its two conditions

For the phase 1 summary, D is a lookup in the phase 1 table, so $D0$ and $Dstep$ become two statements about that table:

- the entry of the solved summary is zero;
- every entry is at most one more than the entry reached from it by any of the eighteen moves.

The table is generated by an OCaml program, and it gives the exact distance. The proof does not need that. The two statements above are all it checks, and they are enough. Together they make the estimate a lower bound on the moves still needed, which is the condition for cutting a branch. A table of zeros would pass both. It would prune nothing and the search would run for ever, but the answer would still be right. So the generator is not trusted. It writes the table out as Rocq source, and the two statements are checked on it afterwards. The entries do not depend on each other. So the second check is cut into slices, one file each, and the slices are checked at the same time. It takes ten minutes. Every timing in this note is measured on the same machine, the **reference machine**: a dual-socket Intel Xeon E5-2667 at 2.9 GHz, twelve cores, twenty-four threads, 62 GB of memory.

5 Optimising

What follows makes the tree smaller or the run cheaper.

5.1 Removing redundant moves

Many words lead to the same position, and the search does not have to try them all. Two ideas say which ones may be left out. The first is repetition. After a U we do not try U , $U2$ or U' . $U \ U$ is $U2$, so the word is shorter, and shorter words are covered at a smaller depth. That leaves fifteen moves instead of eighteen. Opposite faces give a weaker version of the same argument. $U \ D$ and $D \ U$ give the same position, so we keep only one of the two orders. We play the top, right or front face first. This is what we call the **order convention**. From the third move on it leaves **twelve** moves after a turn of the top, right or front face, and **fifteen** after a turn of the bottom, left or back one. The second idea is symmetry. The superflip is unchanged by all 48 relabellings of the cube. So we need to explore only the turns of one face for the first move. We choose arbitrarily the top one, and again by symmetry we only have to consider U and $U2$, since U' is the symmetric of U . The two ideas collide at the second move. After U , repetition removes U , $U2$ and U' , which leaves fifteen. The order convention would remove D , $D2$ and D' as well and leave twelve, but it may not be used here. The first move is already fixed to the top face, and turning the cube upside down takes $D \ U$ back to $U \ D$. So the bottom face stays, and the fifteen second moves after U include $U \ D$, $U \ D2$ and $U \ D'$.

The search is then parallelised at depth two. Two first moves times fifteen second moves is thirty **prefixes**. Each is searched on its own to depth 17. They are packed one file per second move, `Runp1_03.v` to `Runp1_17.v`. The two whose second move turns the bottom face keep fifteen branches where the others keep twelve, so they run far longer. Each of those two is split into two files,

Runp1_09a.v and Runp1_09b.v, Runp1_11a.v and Runp1_11b.v, which balances the load and makes **seventeen files** in all. Our own OCaml program dropped the bottom-face moves, so it searched 24 prefixes where it had to search 30. It ran for hours and gave the answer we expected. The error came out only when the cut had to be proved in Rocq, and the proof could not be written. A cut that is too greedy does not make a search fail. It makes it faster, and it makes it agree with you.

5.2 The effective representation

To run the search we need an effective representation of its objects: of a position first of all, and of the permutations that move it. The objects are few: a position, a move, the summary of a position, and the table of distances. A position is a permutation of the 48 stickers. **Table.v** presents it by its image table, the list of 48 numbers saying where each sticker goes. Not every list of numbers is such a table, and **tab_ok** is the test: the list has 48 entries, each of them below 48, and no entry appears twice. Two permutations are composed by reading one table through the other. To find where sticker i ends up, take entry i of the first table, then use that number as the index into the second. A move is a table of the same shape, so playing a move on a position is that same reading. On four stickers instead of 48, if the first table is (1, 2, 3, 0) and the second is (0, 2, 1, 3), the product is (2, 1, 3, 0). Entry 0 of the first is 1, entry 1 of the second is 2, so the product sends sticker 0 to sticker 2. **Tsearch.v** runs the search of **Search.v** on tables. Next come machine integers, 63 bits wide, and **persistent arrays** of them [8]. **Tabi.v** carries the tables as arrays of machine integers. **ti2t** reads such an array back as the list it stands for, and **tabi_ok** is **tab_ok** of that list. Each operation has a lemma saying that the bridge may be crossed either way round.

```
Lemma ti2t_comp a b :
  tabi_ok a -> tabi_ok b ->
  ti2t (comp_tabi a b) = comp_tab (ti2t a) (ti2t b).
```

From there on a position is 48 machine integers and a summary is two. The phase 1 table is an array of arrays, with fifteen four-bit entries to a 63-bit machine integer. A Rocq array holds at most 4 194 303 entries. The table needs far more, so it is cut into chunks of two million words.

The superflip itself goes down that chain. As a permutation it is a product of twelve two-cycles, one for each flipped edge, (1 33), (3 9), (4 25) and so on. **Moves.v** turns those cycles into the image table **sftab**, and the table into the array **sfti** the search starts from. Each step has its lemma:

```
Lemma sftabE : superflip = pt 47 sftab.
Lemma sftiE : superflip = pt 47 (ti2t 47 sfti).
```

pt 47 is the permutation a table stands for, so both say that what runs is still the superflip. Its summary is read off the same table, the corner twist by **ctwistt** and the flip-and-slice value by **coordt**. The estimate at the root of the search is then one expression:

```
Dpli (ctwistt sftab) (coordt sftab)
```

The summary is (0, 15 732 735). The superflip leaves the corners alone, so the twist is zero, and the second number carries the twelve flipped edges and the four slice slots. The lookup goes through the fold to a four-bit field of one 63-bit integer, and the value there is **ten**. At the root the search therefore knows that at least ten moves are needed, and it has nineteen to spend, so nothing is cut there.

Functions are treated the same way. A function on a finite domain is computed once and written into a table. The action of a move on a summary, the rank of a summary and the symmetry used by the fold are all tables, not computations. The search reads them where the mathematics applies a function. Each of these tables is checked in Rocq, like the table of distances.

The search itself goes the same way. **Fast.v** holds it twice. **searchz3** is the abstract one, written on the objects of the last section. **searchz3n** is what runs, on machine integers and persistent arrays. **FastP.v** ties the two:

```

Lemma searchz3nE T d a p :
  (d <= 63)%N -> fsmoveC -> (p < 7)%N ->
  tabi_ok 47 a -> cubti a -> twP3 a ->
  searchz3n T d (of_nat d) a [::] (init3 a) p
  = searchz3 T d a (init3 a) p.

```

The hypotheses say that the depth fits in a machine integer, that the move table passed its check, and that the array is a well-formed position. Between the two there are seven versions. Each is proved equal to the one before, and together they are 11.9 times faster on one piece at depth 14. The middle versions may be written in any way. The answer stays the one the abstract search gives.

5.3 Two uses of symmetry

The first relabels the position. Rotating the whole cube about a corner axis gives the same position seen differently, and its summary is then another entry of the same table. Each of the three views therefore gives a lower bound on the number of moves left, and so does the largest of the three. That costs three lookups at a position instead of one and buys a sharper cut and a smaller tree. Cube solvers do this all the time, Kociemba’s included. What is new here is the proof that the three views are legitimate.

The second relabels the table. The summary is built around the up-down axis: the twist records where each corner’s up-or-down sticker sits, the slice where the four edges between the top and bottom faces are. Sixteen of the 48 relabellings keep that axis and turn one summary into another, and they sort the 1 013 760 flip-and-slice values into **64 430 families**, a factor of **15.73**. Two values in one family are the same distance from solved, so one entry per family is enough. The estimate is therefore unchanged, and only the table shrinks. In the code the change is one definition. The lookup was

```

Definition Dpli (tw x : int) : int := plget (plidx tw x).

```

and it became

```

Definition Dplri (tw r : int) : int :=
  plget (plfoldi (frep r) (twsym tw (fsym r))).

```

`frep r` is the family’s representative, `fsym r` the symmetry that carries the value to it, and `twsym tw (fsym r)` the twist carried through that same symmetry. Three more reads at every lookup, into a table 15.73 times smaller.

Symmetry-reduced tables are standard in cube solvers. What the development adds is a proof that the folded table still passes **D0** and **Dstep**, and that is all it has to prove. Conditions demanding true distances would have needed a proof that the fold preserves them. That is a harder statement, about the sixteen symmetries and about what sharing an entry between two summaries does. The check is run on the folded table as it was on the flat one, and it is the same check. The fold costs the search 1.61 times at depth 16 and pays everywhere else. A search worker drops from 4.15 GB to **0.85 GB**, so all the pieces run at once instead of in two waves. Checking the table drops from about 5.4 processor hours to **1.35**.

6 The development and its cost

The statement proved at the top of the chain is

```

Theorem superflip_plfar_real : superflip \notin ball Sset pldepth.

```

In words, the superflip is not within `pldepth` moves of the solved cube, where one script sets the depth before the run. It has **no hypotheses left**, and nothing in the chain is admitted: asking Rocq what the proof assumes reports only the primitives of its machine-integer and array interface. At depth 19 it says that the superflip cannot be solved in 19 moves, and two lines in **Diam20.v** turn that into **God’s number ≥ 20** , after checking that the searches really were run at 19.

It is forty-six hand-written files, 12 725 lines about the cube and 1 308 more of machine-integer toolbox, beside 156 000 lines of generated tables in the repository and 165 MB of them too big to store. The cube and its symmetries are `Rubik333.v`, `Sym.v` and `Ball.v`, the abstract search `Search.v` and `Coord.v`, the representations `Table.v` and `Tabi.v`, the summary and its table `Coordfs.v` and `Phase1.v`, and the search on the real data `Farp1.v`, `Fast.v`, the seventeen pieces `Runp1_03.v` to `Runp1_17.v` and `Diam20.v`. The checks live in certificate files of their own, each behind its own `Qed`, and `README.md` lists everything with the scripts that run it.

Building the tables costs the same whatever radius is searched afterwards. Measured end to end from a clean tree on the reference machine:

	wall clock	processor time
emitting the tables and compiling them to native code	17 min 21	52 min 13
the coordinate and summary tables	22 min 46	21 min 15
the search and the estimate, over no data at all	11 min 02	30 min 45
the four certificates for the move and distance tables	56 s	2 min 27
the fold: twelve checks and twenty-seven slices	9 min 52	47 min 57
the shape check against the dummy table	18 s	16 s
in total	1 h 02	2 h 35

The second line runs on one core, so more cores do not help. The first and fifth are mostly the OCaml compiler turning a table into native code, and together they take 100 of the 155 processor-minutes.

The search itself visits 146 065 078 152 positions, from 5 575 767 076 in the smallest piece to 10 554 835 820 in the largest, and the tree grows by 12.22 from one level to the next between depths 17 and 19. We measured the run twice, once before the fold and the two reductions and once after. It is the same theorem both times.

radius 19, search depth 17	before	after
pieces	18	17
workers	9	18
memory per worker	4.15 GB	0.85 GB
wall clock	11 h 13	6 h 36
processor time	85 h 11	87 h 36

The fold halves the wall clock and leaves the processor time where it was. What it buys is memory: at 0.85 GB a worker, all seventeen pieces run at once, where 4.15 GB allowed only nine. So the proof costs 87 processor-hours and one night, and that is a measured cost.

7 Counting in quarter turns

We have proved that God's number is at least 20. We now count the moves differently: in quarter turns there are twelve moves, the six faces one way and the same six back, and a half turn is two moves. The answer in that count is **26** (cube20.org), and we prove the lower half: one position cannot be solved in 25 quarter turns.

7.1 The position, and 25 down to 24

The superflip is only 24 quarter turns from solved, so it is not far enough. Reid posted a better position to the Cube-Lovers list in August 1998 [9]: the superflip composed with four spot, the four-spot pattern with the superflip on top of it. In the following, we call it **superflip4**.



Figure 8: The four-spot, and superflip4.

Superflip4 is 26 quarter turns from solved. Here is the word given for it, with the half turns written out:

$$U U D D L F F U' D R R B U' D' R L F F R U D' R' L U F' B'$$

The searches only need to go to depth 24. A quarter turn is five four-cycles of the 48 stickers, so it is an odd permutation and each move changes the parity of the position. Superflip4 is even, like the solved position, so every word for it has even length.

7.2 The six prefixes

Every shortest word for superflip4 can be rewritten, at the same length, so that it begins with one of six prefixes [9]:

$$\begin{array}{ccc} R U & R' U D & R' U F' \\ R' U R' & R' U B' & R' U L' \end{array}$$

The searches start from those six. In order to prove this fact, we split the twelve quarter turns into two sets:

$$\mathcal{A} = \{ U, U', D, D' \}, \quad \mathcal{C} = \{ R, R', F, F', L, L', B, B' \}.$$

A word made only of turns from $\mathcal{A}$ leaves the four middle edges untouched, and superflip4 has them flipped, so no such word gives superflip4. A word made only of turns from $\mathcal{C}$ never flips an edge, and superflip4 has every edge flipped, so no such word gives it either. Any word for superflip4 therefore uses turns from both sets, so somewhere in it a two-letter subword has one turn from each set.

Each of the six prefixes starts with a turn from $\mathcal{C}$ followed by a turn from $\mathcal{A}$, so we want a word for superflip4 that starts with such a subword. Recall the word we gave for it:

$$U U D D L F F U' D R R B U' D' R L F F R U D' R' L U F' B'$$

Reading along it, four two-letter subwords have a turn of $\mathcal{C}$ followed by a turn of $\mathcal{A}$, and five have them the other way round. We take the first of the four, at the seventh and eighth turns, F then U' , and call that subword q . A word may have subwords of the second kind only. We would then invert it first: superflip4 is its own inverse, so the inverse is again a word for it, of the same length, and inverting turns every subword of the second kind into one of the first. Here the inverse is

$$B F U' L' R D U' R' F' F' L' R' D U B' R' R' D' U F' F' L' D' D' U' U'$$

with, as expected, five subwords of the first kind and four of the second.

Let us go back to our two-letter subword q . Six turns stand before it. Write x for them, $U U D D L F$, and w for what follows q , so that $xqw = P$, where P is superflip4. Then $qw = x^{-1}P$, and therefore

$$qw(P^{-1}xP) = x^{-1}PP^{-1}xP = P.$$

$P^{-1}xP$ is not longer than x :

$$P^{-1}xP = (P^{-1}UP)(P^{-1}UP)(P^{-1}DP)(P^{-1}DP)(P^{-1}LP)(P^{-1}FP) = U U D D R B.$$

The first step is done: qw followed by those six turns is a word for superflip4 of the same length, beginning with q .

$F\ U'\ D\ R\ R\ B\ U'\ D'\ R\ L\ F\ F\ R\ U\ D'\ R'\ L\ U\ F'\ B'\ U\ U\ D\ D\ R\ B$

The second step renames the faces. A relabelling of the cube that leaves superflip4 unchanged carries a word for it to another word for it, of the same length. Sixteen of the 48 relabellings do that. The superflip is unchanged by all 48, but the four-spot is not: it leaves the top and bottom faces alone and exchanges the colours of the other four in pairs, so it singles out the up-down axis, the line through the centres of the top and bottom faces. A relabelling that moves that axis carries the four-spot to the same pattern about another axis, which is another position. The sixteen that keep the axis are the ones that leave superflip4 alone. They make the letter of q from $\mathcal{A}$ into U , and the letter from $\mathcal{C}$ into R or R' , so q becomes $R\ U$ or $R'\ U$. In our example q is $F\ U'$. The mirror that swaps left and right fixes the front face and reverses the direction of every turn, so $F\ U'$ becomes $F'\ U$. A quarter rotation about the up-down axis then carries the front face to the right one, and $F'\ U$ becomes $R'\ U$.

A word starting with $R\ U$ needs nothing more, which is the prefix of two turns. When it starts with $R'\ U$ we look at the third turn. Six of the possible turns give words that a relabelling or an inversion brings back to the $R\ U$ case, and five do not. Those five are the prefixes of three turns.

7.3 The summary, and its table

The estimate is built as before. We take the summary Reid uses in the quarter-turn count. Against the phase 1 summary of the first bound, it keeps the corner twist, the same 2 187 values, and replaces the edge flips and the slice by where the four middle edges sit with their flips, and by which four corner places hold the top corners.

summary	values	
where the four middle-layer edges sit, each of them the right way round or not	190 080	$= 24 \cdot 22 \cdot 20 \cdot 18$
which four corner places hold the four top corners	70	4 places among 8
how the eight corners are twisted	2 187	$= 3^7$
the three together	29 099 347 200	

Each of the four edges can sit in any free slot, either way round. That gives 24 choices for the first, and two fewer for each of the others, since a slot taken is taken whichever way round the edge in it lies. A summary is the coset of a subgroup H of the cube group, which is why the file names of this section start with H . It is not the subgroup of the first bound, and the summaries are thirteen times as many, 29 billion against 2.2 billion.

The table holds the distance from solved of each of the 29 billion summaries. The number of summaries at each distance agrees with the column published in 1998, and we check that first. The table is then folded. The sixteen symmetries that keep the up-down axis sort the 190 080 edge values into 12 094 families, a factor of 15.72, and one entry is kept per family. That is 883 MB, and 3.86 GB once loaded into the prover.

7.4 The theorem, and its cost

The position is settled both ways in `HFinal.v`:

`Theorem targ_dist : targ \in ball Sq 26 /\ targ \notin ball Sq 25.`

`targ` is superflip4 as the searches meet it, and `ball Sq n` is the set of positions within n quarter turns. The word gives the first half and the six searches give the second. The bound follows:

`Theorem qdiam25 : ~ diam_le Sq 25.`

`diam_le Sq 25` says every position is within 25 quarter turns, and the line says it is not. Rocq reports only the primitives of its machine-integer and array interface. The work is eighteen hand-written

files and 6 008 lines. The argument above is in `HProp2.v`, the search in `HSearch.v`, the sweeps in `HSweep.v`, and the bound in `HAll.v`.

	wall clock	processor time
building the table, in OCaml	9 min 50	1 h 43
the same table as fifty-nine Rocq files	3 h 13	6 h 50
the three sweeps over the move tables		1 min 20
the sweep over the distance table, twelve jobs	2 h 51	31 h 42
the six searches, seventy-two pieces, twelve workers	4 h 00	45 h 54
the whole chain in Rocq	10 h 32	87 h 29

The last row is measured end to end, from a directory where nothing is built. It is not the sum of the others. The OCaml table of the first row is built once by hand, and it is not part of that run. The sweeps and the searches are nearly nine tenths of the cost, and checking the table rather than trusting it costs about two thirds of what the searches cost.

8 One coset of the upper half

So far we have dealt with lower bounds. This last section is an attempt at the real computation, the upper bound. The published proofs do not solve all 43 quintillion positions one at a time. They cut the cube group into the 2 217 093 120 cosets of a subgroup and solve a whole coset at once. One search settles every one of the 19 508 428 800 positions in it. The positions in a coset are not all the same distance from solved. The search shows that none of them is more than 20. A symmetry of the cube carries one coset to another, and the image is solved by the same manoeuvres relabelled, so only one coset per symmetry class is searched. Two things are then needed. The cosets searched must cover every class, and each search must settle its whole coset.

`Canon.v` proves the first, and it is not a computation. As the representative of a class we take its least member, in the order the type of positions already carries. Every class is covered, because a finite set that is not empty has a least member. The file is eighty lines and assumes nothing.

The second is the computation itself. Rokicki, Kociemba, Davidson and Dethridge ran it over 55 882 296 families of cosets. It took about a billion seconds of processor time, more than thirty processor years, given by Google. We cannot repeat that. What we can do is one coset, to see what one costs and whether the pieces are in place. We did the superflip's.

8.1 Cosets

Ten of the eighteen moves generate the subgroup H of Kociemba and Thistlethwaite: $U, U2, U', D, D2, D', R2, L2, F2$ and $B2$. They are the moves that do not change the phase 1 summary: they do not twist a corner, they do not flip an edge, and they keep the four middle edges in the middle layer. A coset of H is the set of positions we get from one position using only these ten moves. A coset is therefore one value of the summary, and there are as many cosets as summaries. Every position of the cube lies in exactly one coset.

	count
positions of the cube	43 252 003 274 489 856 000
cosets	2 217 093 120
positions in a coset	19 508 428 800
cosets we did	1

We already had the superflip from the lower bound, so its coset is a convenient example.

8.2 The coset as one map

A position of a coset is named by three numbers: how the eight top and bottom corners sit, how the eight top and bottom edges sit, and how the four middle edges sit. The map holds one bit for each. The arrangements of the corners go in pairs, and a pair shares one word. The map is then 406 425 600 machine words of forty-eight bits, that is 19 508 428 800 bits.

The search starts at the superflip and plays words, setting the bit of every position of the coset it reaches. When every bit is set the theorem follows. Thirty-two positions are not reached. We give each of them a word of twenty moves by hand in `RowWits.v`. We do not trust those words: `RowWitsChk.v` plays each one back and asks for the solved cube.

8.3 The reused and the new

We reuse nearly everything. The search and the map come over unchanged. So do the tables that step a whole map, the ranking of the three numbers and the replay of the leftover words. The phase one table and its generator come over too, with the cube, the permutations and the machine-integer tools.

Four things are new, and none of them is about searching.

- The rank and the sign of a permutation. Rocq's library has both, but for permutations that do not compute, so we need their effective version on machine integers.
- That a position of a coset and its three numbers determine each other. We had one way only, from a position to its numbers. Without the other way, from the numbers back to a position, the theorem would be about triples of numbers and not about the cube.
- The link between the summary the search carries and the position it stands for. The summary is one machine word, the position forty-eight.
- Checks on the tables, one per file, so that each reports separately.

8.4 The unsound stop

A table that underestimates only cuts less, so cutting a branch on it is sound. Stopping on it is not. Our search used to stop when the table gave zero, and take the position it had reached as one of the coset. A table of zeros underestimates everywhere, so it passes the two conditions. It would stop the search at once, and the theorem would say nothing.

The search now stops on the position. At the bottom it tests the position it is carrying. The proof no longer reads the table there, and the test costs one comparison.

8.5 Folding the map

The map is 40 320 pages of 20 160 groups, one page for each way the eight top and bottom corners can sit. Sixteen renamings of the cube keep the top and bottom faces in place, map the ten generators among themselves, and leave the superflip alone. Two pages related by a renaming hold the same answer, so one page of each family is enough: 2 768 of the 40 320, a factor of 14.6. A level of the search is one pass over the map, so there is 14.6 times less of it to walk. The price is undoing a renaming whenever a kept page is read.

The kept pages go in pairs of their own, and a pair shares one word as it does on the unfolded side: 1 496 words in place of 2 768. Two hundred and twenty-four of the pages are their own partner and use half a word.

The fold has to be proved as well as written. A renaming sends a member of the coset to a member of the coset. Undoing it gives back the position the page stood for. A map sound after one level is sound after the next. That is the largest single part of the coset's proof.

8.6 The files

Sixty-nine files, 17 386 lines. The coset and its members are `Row.v`, `RowMemb.v` and `RowInst.v` the ranking and the moves `Lehmer.v` with the `RowPart`, `RowMove` and `RowTab` groups; the map and the

search `RowMap.v`, `RowRun.v`, `RowSrch.v` and `RowFinal.v` the fold the `RowFold` group; and the two runs the `RowCub` and `RowFoldCub` groups, ending in `RowCubDoneI.v` and `RowFoldCubDoneI.v`. `README.md` lists every one of them with what it does.

8.7 The cost

The coset adds sixty-nine hand-written Rocq files and 17 386 lines to the work above, besides the generated tables.

The search ran twice, over the folded map and over the unfolded one, with the same search in both, and both times it filled the map.

the run	wall clock	processor time	peak memory
over the folded map	5 h 53	5 h 51	–
over the unfolded map	7 h 52	7 h 48	23.3 GB

The fold is worth 1.3 times on the wall clock and 1.3 times on processor time. The run does not follow the size of the map, which is 13 times smaller. Most of the work is the search at the deepest levels, and that is the same tree on both sides. The map sets the memory. The unfolded map is 3.25 GB against 248 MB, and a level reads one map while it writes the other, so the unfolded run needed 23.3 GB of a 62 GB machine. With the garbage collector set to work harder, the same run peaks at 13.9 GB and takes four per cent longer.

Two earlier folded runs say where the time went. Over words of twenty-four bits, holding one corner arrangement each, the run took 6 h 00. The same run with the depth left as a unary numeral instead of a machine integer took 8 h 13, so the unary depth cost 2 h 13.

The phase one table is 2.9 GB of Rocq source and 4.5 GB once checked, 8 h 48 of processor time. It is generated once and shared with the lower-bound work.

The statement has no hypothesis left.

Theorem `real_row_superflip_fold_runi m :`
`m \in H -> superflip * m \in ball Sset 20.`

H is the subgroup above, and a coset is one of its cosets. Every position of the superflip's coset is within twenty moves. Rocq reports only the primitives of its machine-integer and array interface. The unfolded run proves the same statement from its own map.

9 Conclusion

We prove two lower bounds. No position of the cube is solved in 19 face turns, and none in 25 quarter turns. In each case one position is the witness, the superflip for the first and the four-spot with the superflip on it for the second. In each case the proof is a search that comes back empty over a table of estimated distances.

We do not prove that 20 and 26 always suffice. The upper half is a different computation. It is not one search that finds nothing, but two billion searches that must each find everything. We did one of the two billion, and the section above says what it cost.

The three share a trunk. The cube itself, as permutations of the forty-eight facelets, with the eighteen moves and the group they generate. Balls, and what it means for a position to be within d moves. The abstract search, whose contract is that a false answer is a proof. The rule that any summary of a position, together with any table that passes one check, gives an estimate that is never too big. And the tables themselves, held as machine integers in arrays rather than as lists of unary numbers.

What each of the three needed of its own:

- **The twenty face turns.** The phase one summary, which is the edge flips and the slice, its table, and the certificate that checks the table. Three viewing angles of the same search, and seventeen pieces run side by side.
- **The twenty-six quarter turns.** The cut to six prefixes, which is the one piece of the development argued by hand rather than computed, and the parity argument that turns 25 into 24. A second and much larger summary, 29 billion values, with a table of its own and a sweep of its own.
- **One coset.** A coset held as a map of bits rather than as a tree of positions. That needs five new pieces. The rank and the sign of a permutation on machine integers. The link between a position and its three ranks. A pass that steps a whole map one move at a time. The fold by the sixteen renamings. And words given by hand for the positions the search does not reach.

The prover is not far behind a program. The same search written in OCaml is about three times faster. We ran both at radius 19 on the reference machine. The OCaml program visits 146 065 078 152 positions in 26.4 processor-hours, which is 0.65 microseconds a position. Rocq takes 87.6 processor-hours over the same tree, which is 2.16. A factor of **3.3**.

We do not assume that the two walk the same tree. We divide each of the seventeen Rocq pieces by the positions its OCaml counterpart visited. The result is between 1.98 and 2.52 microseconds, over pieces that differ in size by a factor of two. A Rocq search that cut differently anywhere would show as scatter there, and there is none. So the run takes a night because the tree holds 146 billion nodes, not because the prover is slow: in OCaml the same tree still costs 26 processor-hours.

The whole development, counted in hand-written Rocq and leaving out the generated tables, is 37 898 lines. Each line of the table counts what that piece adds to the ones above it.

	files	lines
the superflip, for the twenty face turns	53	14 504
the four-spot, for the twenty-six quarter turns	18	6 008
one coset of the upper bound	69	17 386
in all	140	37 898

This development was written with the help of Claude, Anthropic’s coding assistant.

The sources are at github.com/thery/DoubleCover/code/Rubik, with the note, its figures and Reid’s transcribed post beside them.

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